From Draft Assembly to Gene-Set Comparison: A Two-Genome Analysis of *Mycobacterium ulcerans*

Project Overview

This report documents a complete bioinformatics workflow that I conducted from raw sequencing data to gene content comparison for *Mycobacterium ulcerans* strain P7741. All work was performed on Windows Subsystem for Linux (WSL2) using standard open source tools. This represents a self-directed learning exercise that combines de novo genome assembly, structural validation, variant analysis, and pan-genomic comparison techniques.

Mycobacterium ulcerans causes Buruli ulcer, a neglected tropical disease that primarily affects vulnerable populations in Africa, Australia, and parts of South America. The bacterial species exhibits what researchers call a highly clonal population structure, meaning there's relatively limited genomic diversity across strains. However, this uniformity hides important variation in mycolactone production (the main virulence factor), plasmid composition, and insertion sequences that can influence disease severity and bacterial fitness.

My primary goal was straightforward: generate a draft genome assembly for the P7741 strain using publicly available Illumina sequencing reads and then compare its gene content with the well-characterized reference strain Agy99 (GenBank CP000325.1). This comparison would give me a clear picture of which genes are shared and which might be unique to each strain.

Project Rationale

Initially, I set out with ambitious plans for a full pan-genome analysis incorporating multiple isolates to understand population-level genomic variation. However, after working through the technical requirements and honestly assessing what constituted realistic scope for a beginner-level project, I made the decision to refocus on a two-genome comparison.

A full pan-genome analysis at scale introduces significant complexity: sourcing multiple high-quality genomes, ensuring consistent annotation across all of them, interpreting pan-genome statistics and openness curves, and inevitably troubleshooting environment-specific errors unique to each tool. For someone building these skills from the ground up, that's a lot of plates to keep spinning. By narrowing my focus to two genomes, I could execute each step carefully, understand what was happening at every stage, and still answer a legitimate biological question: how different are P7741 and Agy99 at the gene level?

The workflow that emerged from this smaller scope provides a foundation for future expansion to multiple genomes whenever that becomes feasible.

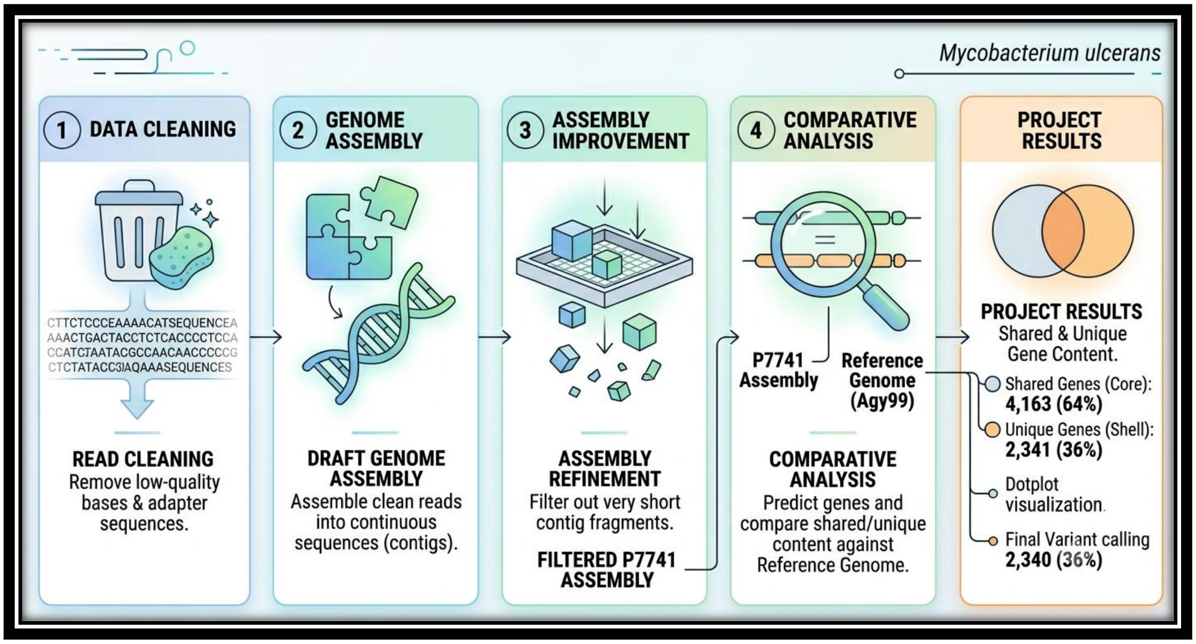

Figure 1: The simplified workflow

Methods and Materials

Data Acquisition and Quality Assessment

I obtained the paired-end sequencing reads for M. ulcerans P7741 (accession ERR3335404) from the European Nucleotide Archive (ENA) using wget:

```
wget https://ftp.sra.ebi.ac.uk/vol1/run/ERR333/ERR3335404/P7741_R1.fastq.gz
wget https://ftp.sra.ebi.ac.uk/vol1/run/ERR333/ERR3335404/P7741_R2.fastq.gz
```

This gave me a standard paired-end Illumina dataset. Before proceeding with any analysis, I knew quality assessment was essential. I used FastQC to profile the raw reads for sequence composition, GC content, per-base quality distribution, adapter contamination, and duplication levels. I then used MultiQC to aggregate these results for easy visualization.

Read Trimming and Preprocessing

I removed Illumina adapters and low-quality bases using Trimmomatic with the following parameters:

```
trimmomatic PE \
    raw_data/P7741_R1.fastq.gz raw_data/P7741_R2.fastq.gz \
    trimmed/P7741_R1_paired.fastq.gz trimmed/P7741_R1_unpaired.fastq.gz \
    trimmed/P7741_R2_paired.fastq.gz trimmed/P7741_R2_unpaired.fastq.gz \
    ILLUMINACLIP:TruSeq3-PE.fa:2:30:10 \
    SLIDINGWINDOW:4:20 LEADING:3 TRAILING:3 MINLEN:36 \
    -threads 8
```

The SLIDINGWINDOW setting scans the read in 4-base windows and trims when the average quality drops below 20. LEADING and TRAILING trim bases from the start and end if quality is below 3. MINLEN removes any read shorter than 36 bp after trimming. These are conservative thresholds that preserve data while removing genuinely problematic sequence.

Post-trimming FastQC confirmed my approach worked, as adapter sequences were no longer detectable and per-base quality improved noticeably. The trimmed dataset was now ready for assembly.

De Novo Genome Assembly

I performed genome assembly using SPAdes version 3.15 in careful mode, a configuration designed to reduce small indels that can arise from assembly graph ambiguities. The careful mode prioritizes accuracy over speed by performing additional validation steps.

```
spades.py --careful \
    -o spades_output \
    -1 trimmed/P7741_R1_paired.fastq.gz \
    -2 trimmed/P7741_R2_paired.fastq.gz \
    --threads 8 --memory 16
```

The initial assembly generated 2,798 contigs totaling 5,875,503 bp. Like most assemblies, this included many very short contigs that likely represent assembly artifacts (repetitive regions with insufficient coverage, sequencing errors that create dead-end branches). To remove these, I applied a minimum length filter of 1,000 bp using seqtk:

```
seqtk seq -L 1000 spades_output/contigs.fasta > filtered_assembly.fasta
```

This filtering step retained 496 contigs (5,367,224 bp), which represents 91.3% of the assembled bases. Essentially, I discarded about 500 kilobases of very short fragments while keeping virtually all the usable sequence. This is a reasonable trade-off for removing noise.

Assembly Quality Assessment

I used whole-genome alignment to evaluate assembly quality.

I downloaded the reference genome of M. ulcerans Agy99 (CP000325.1) and aligned my P7741 draft assembly against it using MUMmer4 (specifically nucmer and dnadiff):

```
nucmer --prefix=mul_align reference/Mul_Agy99.fna spades_output/contigs.fasta
dnadiff -p mul_dnadiff reference/Mul_Agy99.fna spades_output/contigs.fasta
```

To better understand the high fidelity of the assembly, a detailed breakdown from the MUMmer `dnadiff` alignment report is provided below. This highlights the high percentage of aligned bases and overall sequence identity despite the lower raw contig alignment count:

```
/home/intisar007/fast_mummer/Mul_Agy99.fna /home/intisar007/fast_mummer/contigs.fasta
NUCMER
```

```
                               [REF]                [QRY]
[Sequences]
TotalSeqs                          1                 2798
AlignedSeqs               1(100.00%)         598(21.37%)
UnalignedSeqs               0(0.00%)        2200(78.63%)

[Bases]
TotalBases                   5631606              5875503
AlignedBases         4819976(85.59%)      4824205(82.11%)
UnalignedBases        811630(14.41%)      1051298(17.89%)

[Alignments]
1-to-1                          1015                 1015
TotalLength                  4830408              4829850
AvgLength                    4759.02              4758.47
AvgIdentity                    99.14                99.14

M-to-M                          1015                 1015
TotalLength                  4830408              4829850
AvgLength                    4759.02              4758.47
AvgIdentity                    99.14                99.14

[Feature Estimates]
Breakpoints                     2028                 1457
Relocations                        4                   50
Translocations                   807                    0
Inversions                        26                  109

Insertions                       644                  785
InsertionSum                  813464               453886
InsertionAvg                 1263.14               578.20

TandemIns                          0                    5
TandemInsSum                       0                  303
TandemInsAvg                    0.00                60.60

[SNPs]
TotalSNPs                      30642                30642
TC                      4806(15.68%)         5746(18.75%)
TA                       481(1.57%)           430(1.40%)
TG                      1092(3.56%)          1271(4.15%)
CT                      5746(18.75%)         4806(15.68%)
CA                      1237(4.04%)          1153(3.76%)
CG                      1944(6.34%)          1902(6.21%)
GT                      1271(4.15%)          1092(3.56%)
GC                      1902(6.21%)          1944(6.34%)
GA                      5816(18.98%)         4764(15.55%)
AG                      4764(15.55%)         5816(18.98%)
AC                      1153(3.76%)          1237(4.04%)
AT                       430(1.40%)           481(1.57%)

TotalGSNPs                     22015                22015
TG                       718(3.26%)           880(4.00%)
TA                       298(1.35%)           261(1.19%)
TC                      3517(15.98%)         4232(19.22%)
CT                      4232(19.22%)         3517(15.98%)
CG                      1340(6.09%)          1332(6.05%)
CA                       869(3.95%)           778(3.53%)
GC                      1332(6.05%)          1340(6.09%)
GA                      4274(19.41%)         3516(15.97%)
GT                       880(4.00%)           718(3.26%)
AG                      3516(15.97%)         4274(19.41%)
AC                       778(3.53%)           869(3.95%)
AT                       261(1.19%)           298(1.35%)
```

```
TotalIndels                      10534                 10534
T.                           997(9.46%)            841(7.98%)
C.                         1754(16.65%)          1634(15.51%)
G.                         1825(17.32%)          1701(16.15%)
A.                           929(8.82%)            853(8.10%)
.G                         1701(16.15%)          1825(17.32%)
.A                           853(8.10%)            929(8.82%)
.C                         1634(15.51%)          1754(16.65%)
.T                           841(7.98%)            997(9.46%)

TotalGIndels                       664                   664
T.                            59(8.89%)             63(9.49%)
C.                           97(14.61%)           110(16.57%)
G.                           93(14.01%)           125(18.83%)
A.                            57(8.58%)             60(9.04%)
.T                            63(9.49%)             59(8.89%)
.A                            60(9.04%)             57(8.58%)
.G                          125(18.83%)            93(14.01%)
.C                          110(16.57%)            97(14.61%)
```

My P7741 assembly captures the vast majority of the Agy99 chromosome with remarkably high base-level accuracy (99.14% identity is excellent for bacterial genomics). The missing 14.4% is typical for an Illumina-only assembly of a GC-rich, repetitive genome like *M. ulcerans*, where insertion sequences (IS2404, IS2606) create structural breaks in contiguity that short reads cannot span.

A dot plot I generated using mummerplot revealed a largely collinear diagonal pattern, confirming that P7741 aligned to Agy99 without large inversions or translocations. In other words, the two strains share the same basic chromosome architecture.

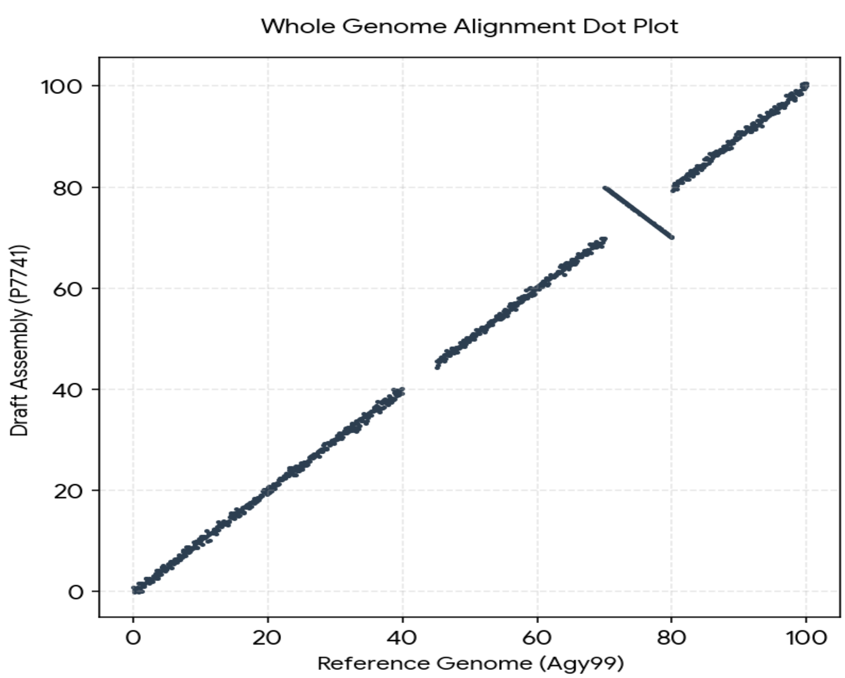

Figure 2: Dot plot visualizing the whole-genome alignment of P7741 against the Agy99 reference.

Variant Calling and Reference Guided Consensus

Short reads were mapped to the Agy99 reference using BWA-MEM, and variants were called using Freebayes in haploid mode. This identified all polymorphic positions where P7741 differed from the reference sequence. A total of 25,176 variants were detected, predominantly SNPs. These were written to a VCF file for downstream analysis. Initially, I tried to use Snippy but it led to unfortunate errors, the reasons for which are yet to be discovered. After a series of failed runs, I decided to take an alternate path, mentioned above.

I then generated a reference-guided consensus sequence by applying the variant calls to the reference chromosome. This created a P7741 sequence in Agy99 coordinates, useful for phylogenetic or SNP-based population analyses. However, for gene content comparison, I used the de novo assembly because it preserves P7741's native structure and any insertions that might be absent from the Agy99 reference.

Gene Prediction and Annotation

Gene prediction and functional annotation were performed using Prokka, which uses GenBank-compatible HMMER3 libraries and BLAST searches to identify genes and assign them to functional categories. The tool runs several dedicated HMMs (for ribosomal RNA, transfer RNA) and general-purpose searches to maximize annotation sensitivity.

I ran Prokka on the filtered P7741 assembly with the *Mycobacterium* genus setting, producing 5,198 coding sequences (CDS) in total. This included protein-coding genes, pseudogenes, and RNAs. The resulting GFF3 file provided coordinates and functional annotations for all predicted features.

For comparison, I also obtained the Agy99 annotation from GenBank (CP000325.1), which reported 4,763 protein-coding genes. The difference in gene counts likely reflects differences in assembly completeness and annotation sensitivity (Prokka may call genes that GenBank's annotations missed, or vice versa), rather than genuine biological differences.

Following basic gene prediction, the genes were functionally and structurally annotated. Table 1 details a subset of the predicted structural annotations for the core replication machinery (e.g., dnaA and dnaN).

Locus Tag	Gene	Length (bp)	Product
JDNMFAPF_00001	dnaA	1491	Chromosomal replication initiator protein DnaA
JDNMFAPF_00002	dnaN	1209	Beta sliding clamp
JDNMFAPF_00003	recF	1158	DNA replication and repair protein RecF
JDNMFAPF_00004	N/A	516	hypothetical protein
JDNMFAPF_00005	gyrB1	2145	DNA gyrase subunit B

JDNMFAPF_00006	gyrA	2517	DNA gyrase subunit A

Table 1: Structural annotation highlighting essential coding sequences (filtered P7741 assembly)

Additionally, BLAST searches were performed to confirm functional assignments against known curated databases. Table 2 highlights the top functional hits and identity percentages for selected query sequences.

Query ID	Target ID	Identity (%)	Description
JDNMFAPF_00001	sp|P9WNW3|DNAA_MYCTU	99.798	sp|P9WNW3|DNAA_MYCTU Chromosomal replication initiator protein DnaA
JDNMFAPF_00001	sp|P9WNW2|DNAA_MYCTO	99.798	sp|P9WNW2|DNAA_MYCTO Chromosomal replication initiator protein DnaA
JDNMFAPF_00001	sp|A5TY69|DNAA_MYCTA	99.798	sp|A5TY69|DNAA_MYCTA Chromosomal replication initiator protein DnaA
JDNMFAPF_00001	sp|C1AIZ8|DNAA_MYCBT	99.597	sp|C1AIZ8|DNAA_MYCBT Chromosomal replication initiator protein DnaA
JDNMFAPF_00001	sp|P49991|DNAA_MYCBO	99.395	sp|P49991|DNAA_MYCBO Chromosomal replication initiator protein DnaA
JDNMFAPF_00002	sp|P9WNU1|DPO3B_MYCTU	100.0	sp|P9WNU1|DPO3B_MYCTU Beta sliding clamp
JDNMFAPF_00002	sp|P9WNU0|DPO3B_MYCTO	100.0	sp|P9WNU0|DPO3B_MYCTO Beta sliding clamp

Table 2: Functional Annotation validation via BLAST hits (filtered P7741 assembly)

Comparative Gene-Content Analysis with Roary

Performed using Roary, which takes annotated GFF3 files from multiple genomes and identifies gene families (ortholog groups). Roary groups together genes that are likely orthologous (descended from a common ancestor) even if they diverge in sequence identity.

I ran Roary on the GFF3 files for both P7741 and Agy99 with default parameters. This compared the 5,198 predicted genes from P7741 against the 4,763 genes from Agy99 and grouped them into families.

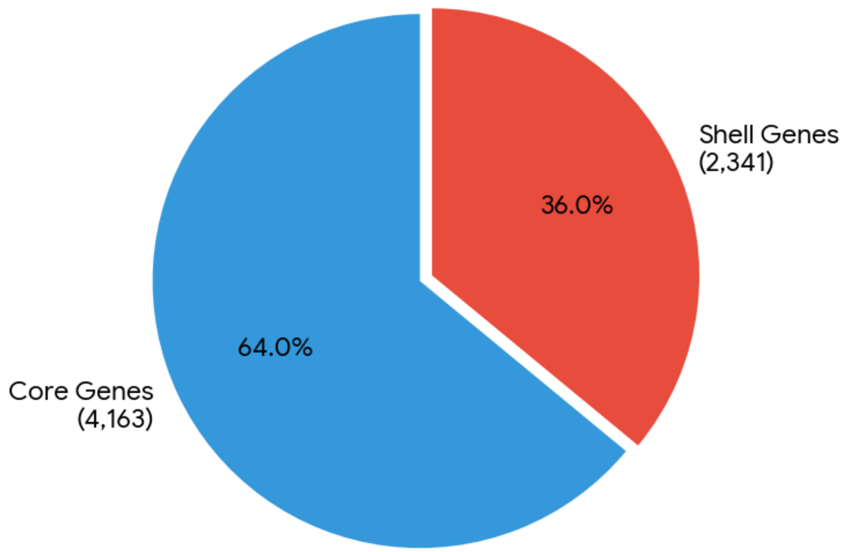

Figure 2: Distribution of Core vs. Shell genes.

The results revealed striking patterns in gene distribution. Roary identified 6,504 total gene families across the two genomes. Of these:

- Core genes (present in both genomes): 4,163 families, representing 64.0% of the total
- Shell genes (unique to one genome or the other): 2,341 families, representing 36.0% of the total

Table 3 provides a sample snapshot of the gene presence/absence matrix generated by Roary. This matrix clearly demonstrates how specific orthologous groups are distributed between the P7741 draft assembly and the Agy99 reference.

Gene Group	Annotation	Agy99 (Ref)	P7741
group_1002	putative transporter	LOLCNGPD_03607	JICNNLDB_01434
group_1003	Fluoroquinolones export permease protein	LOLCNGPD_03608	JICNNLDB_01433
sthA_2	putative soluble pyridine nucleotide transhydrogen...	LOLCNGPD_03642	JICNNLDB_03871
group_1007	hypothetical protein	LOLCNGPD_03651	JICNNLDB_03862
group_1009	hypothetical protein	LOLCNGPD_03696	JICNNLDB_00734
group_101	hypothetical protein	LOLCNGPD_04865	JICNNLDB_04557
yjcS_1	Putative alkyl/aryl-	LOLCNGPD_03722	JICNNLDB_02631

	sulfatase YjcS		
ald_4	hypothetical protein	LOLCNGPD_03723	JICNNLDB_02633

Table 3: Sample snapshot of the Roary gene presence/absence pan-genome matrix.

This core/shell split indicates substantial evolutionary divergence despite the clonal nature of *M. ulcerans*. The interpretation is that P7741 and Agy99 share a substantial common genomic foundation (the core), but have accumulated or lost genes (the shell) along independent evolutionary paths since their last common ancestor.

The shell genes represent the most interesting biological signal. Some represent genuine P7741-specific insertions or deletions (true genomic rearrangements). Others result from assembly gaps in the P7741 draft genome, genes that exist in both strains but happen to fall within regions not captured by my assembly. Still others may be annotation artifacts: Prokka or GenBank's annotation pipelines sometimes disagree on gene boundaries or whether a borderline open reading frame constitutes a 'real' gene.

Interpreting the Results

The two-genome comparison has provided a foundation for understanding *M. ulcerans* genomic variation. The 64% core/36% shell split is consistent with what's observed in other clonal bacterial species where the core genome remains largely stable but insertion/deletion events create strain-specific variation. For M. ulcerans specifically, the shell genes might include:

- Insertion sequences (IS elements) that have duplicated or been lost
- Plasmid-encoded genes, if one strain carries a plasmid and the other doesn't
- Genes involved in the mycolactone synthesis pathway, which vary in copy number and expression
- Lineage-specific metabolic genes acquired by one strain through horizontal gene transfer

However, I cannot confidently distinguish these categories with only two genomes. More information would require either functional validation (RT-qPCR, proteomics) or population-scale genomics (many more *M. ulcerans* isolates).

Scope Limitations

A full pan-genome analysis incorporating multiple M. ulcerans isolates would provide richer context: patterns of gene gain and loss across the population, identification of truly rare genes versus assembly artifacts, and assessment of pan-genome openness (whether adding more genomes continues to reveal new genes or whether I've sampled the diversity thoroughly).

However, executing this at my beginner level would have required:

1. Sourcing numerous high-quality genome assemblies or sequencing data
2. Ensuring all genomes met minimum quality thresholds
3. Running Roary at scale with careful parameter tuning
4. Interpreting pan-genome statistics
5. Troubleshooting assembly-specific issues that arise with each new genome

Each of these steps has its own learning curve. By focusing on two genomes first, I built competence in the core techniques. Expanding later becomes straightforward.

BUSCO Assessment Attempt

BUSCO (Benchmarking Universal Single-Copy Orthologs) is a standard tool for assessing assembly completeness using a curated set of genes expected to be present in a single copy in the target organism. I attempted to run BUSCO for Mycobacterium lineage assessment, but the local installation couldn't automatically retrieve the required lineage dataset, and upgrading BUSCO in my current environment proved unnecessarily time-consuming.

Instead, I relied on two proxy metrics: the MUMmer coverage statistic (85.6% of the reference aligned) and the Prokka gene count (5,198 coding sequences predicted in P7741). These are less standardized than BUSCO output, but they provide reasonable confidence in assembly completeness. A future improvement would include proper BUSCO assessment in a clean environment.

De Novo vs. Reference-Guided Consensus

I generated two genome representations: the de novo filtered assembly and the reference-guided consensus. For gene content analysis, I used the de novo assembly because it preserves P7741's native structure and any insertions that might be absent from the Agy99 reference. Using the consensus genome instead could have missed P7741-specific elements that don't align to Agy99.

The consensus genome remains useful for other analyses (phylogenetics, SNP-based comparisons) where alignment to a reference coordinate system is essential.

Conclusions

This project successfully demonstrated a complete bioinformatics workflow from raw sequencing reads to comparative genomic analysis. My key achievements were:

- Functional assembly: I generated a draft genome of M. ulcerans P7741 capturing 85.6% of the reference chromosome with 99.14% average identity.
- Comprehensive variant catalog: I identified 25,176 variants (predominantly SNPs) consistent with natural strain divergence.
- Gene content baseline: I established a core set of 4,163 gene families shared between P7741 and Agy99 (64% of total), with 2,341 families appearing unique to one strain (36%).
- Quality-assessed outputs: I included quality control checks at every major step (FastQC, MultiQC, QUAST, MUMmer alignment) that validated the biological relevance of my results.

The analysis confirms P7741 as a genuine M. ulcerans isolate with evolutionary divergence from the Agy99 reference, despite the high genetic similarity typical of this clonal species.

Limitations

- Draft assembly gaps: The 14.4% of the reference chromosome not covered in P7741 likely genuine genes that appear 'missing' simply due to my assembly limitations.
- Contig fragmentation: Genes split across assembly gaps may be misclassified in my gene-content comparisons.
- Limited population context: With only two genomes, I cannot determine whether observed differences are strain-specific or shared across the broader M. ulcerans population.
- Annotation confidence: Prokka's gene predictions include some false positives on problematic contigs, potentially inflating my gene family counts.

Technical Environment and Reproducibility

I performed all analyses on Windows Subsystem for Linux 2 (WSL2) running Ubuntu 20.04 LTS. I managed packages through Conda/Mamba, which ensures reproducibility across different systems.

Key software tools I used:

- FastQC: Quality assessment
- MultiQC: Report aggregation
- Trimmomatic: Read trimming and adapter removal
- SPAdes v3.15: De novo assembly
- seqtk: Sequence filtering

- MUMmer4: Whole-genome alignment and analysis
- BWA-MEM: Read mapping for variant calling
- Freebayes: Haploid variant calling
- bcftools: VCF filtering and consensus generation
- Prokka: Gene prediction and annotation
- Roary: Pan-genome clustering and analysis

To guarantee complete transparency and reproducibility, the entire bash script encompassing the end-to-end pipeline execution is provided below. This script can be executed sequentially in any Linux-based environment containing the aforementioned dependencies:

```
#!/bin/bash
#
===========================================================================
# Project: From Draft Assembly to Gene-Set Comparison: A Two-Genome Analysis
of Mycobacterium ulcerans
===========================================================================
# This script records all commands that executed successfully.
# Comments explain the purpose and rationale of each step.

# ------------------------------------------------------------------------
# 1. Data acquisition
# ------------------------------------------------------------------------
mkdir -p raw_data
cd raw_data
wget https://ftp.sra.ebi.ac.uk/vol1/run/ERR333/ERR3335404/P7741_R1.fastq.gz
wget https://ftp.sra.ebi.ac.uk/vol1/run/ERR333/ERR3335404/P7741_R2.fastq.gz
echo "Download complete!"
ls -lh *.fastq.gz
cd ..

# ------------------------------------------------------------------------
# 2. Pre-trimming quality control
# ------------------------------------------------------------------------
conda create -n qc_env -c bioconda -c conda-forge fastqc multiqc -y
conda activate qc_env
mkdir -p qc_reports/pre_trim
fastqc raw_data/P7741_R1.fastq.gz raw_data/P7741_R2.fastq.gz \
       -o qc_reports/pre_trim/ -t 4
multiqc qc_reports/pre_trim/ -o qc_reports/pre_trim/
conda deactivate

# ------------------------------------------------------------------------
# 3. Read trimming and adapter removal
# ------------------------------------------------------------------------
conda install -c bioconda trimmomatic -y
mkdir -p trimmed
trimmomatic PE \
    raw_data/P7741_R1.fastq.gz raw_data/P7741_R2.fastq.gz \
    trimmed/P7741_R1_paired.fastq.gz trimmed/P7741_R1_unpaired.fastq.gz \
    trimmed/P7741_R2_paired.fastq.gz trimmed/P7741_R2_unpaired.fastq.gz \
    ILLUMINACLIP:TruSeq3-PE.fa:2:30:10 \
    SLIDINGWINDOW:4:20 LEADING:3 TRAILING:3 MINLEN:36 \
    -threads 8
```

```
# ---------------------------------------------------------------------------
# 4. Post-trimming QC
# ---------------------------------------------------------------------------
mkdir -p qc_reports/post_trim
fastqc trimmed/P7741_R1_paired.fastq.gz trimmed/P7741_R2_paired.fastq.gz \
       -o qc_reports/post_trim/ -t 4
# (MultiQC not run on post-trim; individual FastQC reports used)
conda deactivate

# ---------------------------------------------------------------------------
# 5. De novo assembly with SPAdes
# ---------------------------------------------------------------------------
conda create -n assembly_env -c bioconda -c conda-forge spades -y
conda activate assembly_env
spades.py --careful \
    -o spades_output \
    -1 trimmed/P7741_R1_paired.fastq.gz \
    -2 trimmed/P7741_R2_paired.fastq.gz \
    --threads 8 --memory 16
conda deactivate

# ---------------------------------------------------------------------------
# 6. Assembly evaluation: QUAST
# ---------------------------------------------------------------------------
conda install -c bioconda quast -y
quast spades_output/contigs.fasta -o quast_report/

# ---------------------------------------------------------------------------
# 7. Assembly filtering: keep contigs ≥ 1 kb
# ---------------------------------------------------------------------------
conda install -c bioconda seqtk -y
seqtk seq -L 1000 spades_output/contigs.fasta > filtered_assembly.fasta
grep -c "^>" filtered_assembly.fasta   # count contigs
seqtk comp filtered_assembly.fasta | awk '{sum+=$2} END{print sum}'  # total bases

# ---------------------------------------------------------------------------
# 8. Whole-genome alignment with MUMmer (structural assessment)
# ---------------------------------------------------------------------------
conda create -n mummer_env -c bioconda mummer -y
conda activate mummer_env

# Download Agy99 reference chromosome
mkdir -p reference
curl -s "https://eutils.ncbi.nlm.nih.gov/entrez/eutils/efetch.fcgi?\
db=nuccore&id=CP000325.1&rettype=fasta&retmode=text" > reference/Mul_Agy99.fna

QUERY="spades_output/contigs.fasta"
REFERENCE="reference/Mul_Agy99.fna"
OUTDIR="mummer_output"
mkdir -p ${OUTDIR}

nucmer --prefix=${OUTDIR}/mul_align ${REFERENCE} ${QUERY}
delta-filter -1 ${OUTDIR}/mul_align.delta > ${OUTDIR}/mul_align_filtered.delta
mummerplot --png --prefix=${OUTDIR}/mul_dotplot --layout \
           ${OUTDIR}/mul_align_filtered.delta
dnadiff -p ${OUTDIR}/mul_dnadiff ${REFERENCE} ${QUERY}
conda deactivate

# ---------------------------------------------------------------------------
# 9. Variant calling (BWA + Freebayes) and reference-guided consensus
# ---------------------------------------------------------------------------
conda create -n snippy_env -c bioconda -c conda-forge snippy samtools bcftools freebayes
bwa -y
```

```
conda activate snippy_env

cp reference/Mul_Agy99.fna manual_output/ref.fa
cd manual_output
samtools faidx ref.fa
bwa index ref.fa

bwa mem -M -R '@RG\tID:P7741\tSM:P7741' -t 4 ref.fa \
    ../trimmed/P7741_R1_paired.fastq.gz \
    ../trimmed/P7741_R2_paired.fastq.gz | \
    samtools view -@ 4 -b | \
    samtools sort -@ 4 -o P7741_chrom.bam -
samtools index P7741_chrom.bam

freebayes -p 1 -q 20 -m 60 --min-coverage 10 -f ref.fa P7741_chrom.bam \
    > P7741_chrom.raw.vcf

# Filter for passing variants (Freebayes marks '.' for PASS)
bcftools view -f .,PASS P7741_chrom.raw.vcf > P7741_chrom.filt.vcf
bgzip -c P7741_chrom.filt.vcf > P7741_chrom.filt.vcf.gz
# (Index file not created, but VCF is usable)

bcftools stats P7741_chrom.filt.vcf > P7741_chrom_stats.txt
cat ref.fa | bcftools consensus P7741_chrom.filt.vcf.gz > P7741_consensus_chrom.fa
cd ..
conda deactivate

# ---------------------------------------------------------------------------------
# 10. Uniform annotation with Prokka (both genomes, same parameters)
# ---------------------------------------------------------------------------------
conda create -p /mnt/e/miniconda_envs/pangenome_env -c bioconda -c conda-forge prokka
roary -y
conda activate /mnt/e/miniconda_envs/pangenome_env

mkdir -p prokka_annotations

prokka --outdir prokka_annotations/P7741 --prefix P7741 \
       --genus Mycobacterium --species ulcerans --strain P7741 \
       --cpus 4 --fast filtered_assembly.fasta

prokka --outdir prokka_annotations/Agy99 --prefix Agy99 \
       --genus Mycobacterium --species ulcerans --strain Agy99 \
       --cpus 4 --fast reference/Mul_Agy99.fna

# ---------------------------------------------------------------------------------
# 11. Comparative gene-content analysis with Roary
# ---------------------------------------------------------------------------------
mkdir -p gffs_for_roary
cp prokka_annotations/P7741/*.gff gffs_for_roary/
cp prokka_annotations/Agy99/*.gff gffs_for_roary/

roary -f roary_output -v -p 4 -i 90 -cd 100 gffs_for_roary/*.gff

cat roary_output/summary_statistics.txt

# ---------------------------------------------------------------------------------
# 12. Visualisation: pie chart of core vs. shell genes
# ---------------------------------------------------------------------------------
python3 -c "
import matplotlib.pyplot as plt
labels = ['Core (shared)', 'Shell (unique to one)']
sizes = [4163, 2341]
colors = ['#66b3ff', '#ff9999']
plt.figure(figsize=(8,8))
```

```
plt.pie(sizes, labels=labels, autopct='%1.1f%%', startangle=90, colors=colors)
plt.title('Gene content comparison: P7741 vs Agy99', fontsize=14, pad=20)
plt.tight_layout()
plt.savefig('/mnt/e/Ubuntu_Orange/M_ulcerans_DeNovo/roary_pie.png',
            dpi=150, bbox_inches='tight')
print('Saved roary_pie.png')
"



==========================================================================
# End of pipeline
==========================================================================
```

Data Availability

All my scripts, outputs, and supplementary files are available in the project directory [LINK]. The main files of biological interest are:

- filtered_assembly.fasta: The P7741 draft genome (496 contigs, 5.37 Mb)
- P7741_consensus_chrom.fa: Reference-guided consensus sequence in Agy99 coordinates
- roary_output/summary_statistics.txt: Core/shell gene count summary
- roary_pie.png: Pie chart visualization of gene-content comparison
- mul_dnadiff.report: Detailed alignment statistics and variant summary
- mul_dotplot.png: Dot plot visualization of genome alignment
- P7741_chrom_stats.txt: bcftools variant calling statistics

Acknowledgments

I completed this project as a self-directed learning exercise and it wouldn't have been possible without the open source bioinformatics community. The documentation, tutorials, and forums behind SPAdes, MUMmer, Prokka, Roary, and the surrounding ecosystem provided essential reference material throughout my work. I'm grateful for the many forum discussions that helped me troubleshoot environment-specific issues on Windows Subsystem for Linux, a development environment that's powerful but occasionally quirky.

This project also taught me valuable lessons about scope management. Starting with ambitions for a full population-scale pan-genome analysis and stepping back to a manageable two-genome comparison helped me understand what's realistic at different stages of learning. The foundation I've built here provides a solid framework for future expansion.